

Chapter 3. Dual-spin Stabilization

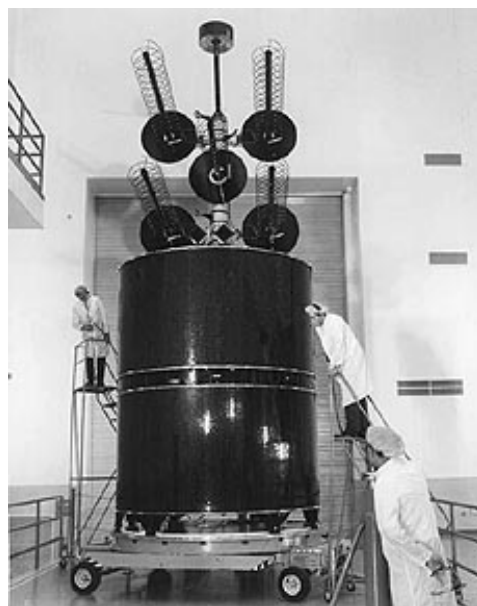
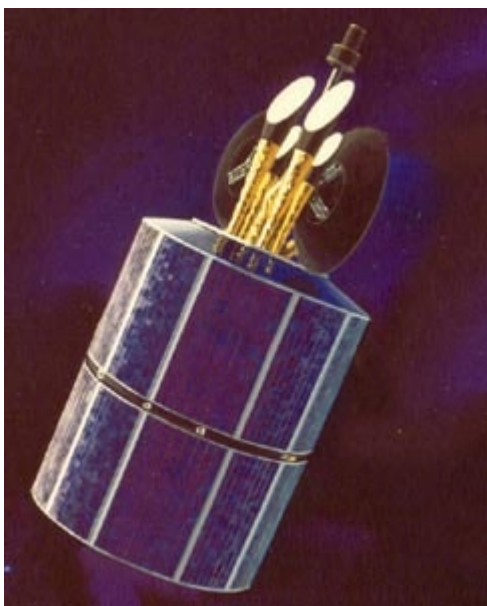
For spin stabilization to be useful in the presence of energy dissipation, the satellite has to be oblate (disk shaped). But many satellites have to be prolate because they have to fit in the Launch vehicles. These satellites can be stabilized using the dual-spin concept.

Dual-spin stabilization of attitude control is often used when the ability to point instrument or provide a platform with good spin stabilization is required.

Advantages: The spacecraft is stable about the transverse axes for long periods of time. Despun platform provides pointing for antenna and instruments.

Disadvantages: The despun drive assembly (motor, bearings, slip rings) is expensive. Complex nutation dynamics. The large energy dissipation of the despun section requires expensive requirement

Examples: Intelsat IV, TACSAT



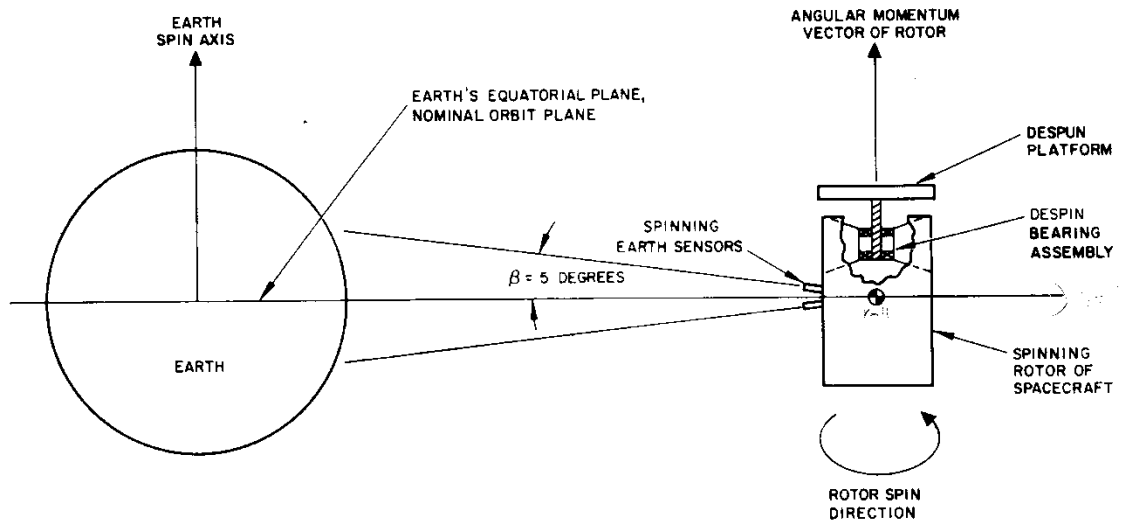
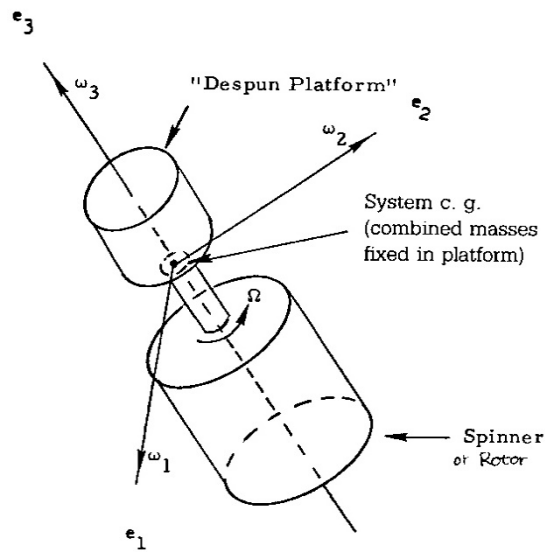


Figure 3.1 Dual-spin concept.

Simplified dual-spin system dynamics



e_1, e_2, e_3 : body frame axis

Let the components of the system angular momentum along the $\hat{e}_i$ axis.

$$\begin{aligned} h_1 &= A\omega_1 \\ h_2 &= A\omega_2 \\ h_3 &= C_1\omega_3 + C_2\Omega \end{aligned} \tag{1}$$

where A ; total transverse moment of inertia

C_1, C_2 ; platform and spinner inertias along the e_3 axis, respectively

ω_i ; inertial angular velocities of the platform

Ω ; inertial angular velocity of rotor

All the components are expressed in the body-frame.

The Euler's equation

$$\begin{aligned} \dot{\vec{h}} + \vec{\omega} \times \vec{h} &= \vec{T} \\ \left. \begin{aligned} \rightarrow \dot{h}_1 + \omega_2 h_3 - \omega_3 h_2 &= T_1 = 0 \\ \dot{h}_2 + \omega_3 h_1 - \omega_1 h_3 &= T_2 = 0 \\ \dot{h}_3 + \omega_1 h_2 - \omega_2 h_1 &= T_3 = 0 \end{aligned} \right\} \text{torque free motion} \end{aligned} \quad (2)$$

Substituting Eq. (1) into Eq. (2), assuming $\dot{\Omega} = 0$ (constant spin) gives

$$\begin{aligned} A\dot{\omega}_1 + (C_1\omega_3 + C_2\Omega)\omega_2 - A\omega_2\omega_3 &= 0 \\ A\dot{\omega}_2 + A\omega_1\omega_3 - (C_1\omega_3 + C_2\Omega)\omega_1 &= 0 \end{aligned} \quad (3)$$

$$\begin{aligned} C\dot{\omega}_3 + A\omega_1\omega_2 - A\omega_1\omega_2 &= 0 \\ \rightarrow \dot{\omega}_1 + \left(\frac{C_1\omega_3 + C_2\Omega - A\omega_3}{A} \right) \omega_2 &= 0 \\ \dot{\omega}_2 - \left(\frac{C_1\omega_3 + C_2\Omega - A\omega_3}{A} \right) \omega_1 &= 0 \\ C\dot{\omega}_3 = 0 \quad \rightarrow \quad \omega_3 = \text{const.} \end{aligned} \quad (4)$$

$$\begin{aligned} \rightarrow \dot{\omega}_1 + \lambda\omega_2 &= 0 \\ \dot{\omega}_2 - \lambda\omega_1 &= 0 \end{aligned} \quad (5)$$

$$\text{where } \lambda = \frac{C_1\omega_3 + C_2\Omega - A\omega_3}{A}$$

$$\begin{aligned} \rightarrow \ddot{\omega}_1 + \lambda\dot{\omega}_2 &= 0 & \rightarrow \ddot{\omega}_1 + \lambda^2\omega_1 &= 0 \\ \ddot{\omega}_2 - \lambda\dot{\omega}_1 &= 0 & \ddot{\omega}_2 + \lambda^2\omega_2 &= 0 \end{aligned}$$

The solution of Eq. (5) yields,

$$\begin{aligned} \rightarrow \omega_1 &= \omega_1(0)\cos \lambda t \\ \omega_2 &= \omega_2(0)\sin \lambda t \end{aligned}$$

where $\omega_1(0)$ and $\omega_2(0)$ are the initial values of ω_1 and ω_2 respectively, λ represents the frequency of ω_1 and ω_2 .

The transverse angular velocity is

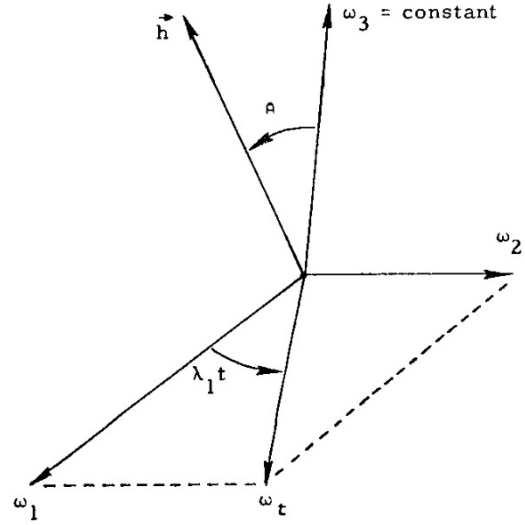
$$\omega_t = (\omega_1^2 + \omega_2^2)^{1/2}$$

$\vec{h}$ is the fixed angular momentum vector

$$\begin{aligned} h &= (h_1^2 + h_2^2 + h_3^2)^{1/2} \\ &= \sqrt{(A\omega_t)^2 + (C_1\omega_3 + C_2\Omega)^2} \\ &= \text{constant } (\text{torque-free}) \end{aligned}$$

The angle θ is nutation angle

$$\tan \theta = \frac{h_t}{h_3} = \frac{A\omega_t}{C_1\omega_3 + C_2\Omega}$$



Dual-spin system stability considerations

The kinetic energy E_k is conserved if the system is a perfectly rigid.

$$E_k = \frac{1}{2} \vec{\omega} \cdot \vec{h}$$

$$= \frac{1}{2} [\vec{\omega}_p \cdot \vec{h}_p + \vec{\omega}_r \cdot \vec{h}_r]$$

$$\text{assume } \vec{\omega}_p = \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix}, \quad I_p = \begin{bmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C_1 \end{bmatrix}$$

$$\Rightarrow \vec{h}_p = \begin{Bmatrix} A\omega_1 \\ A\omega_2 \\ C_1\omega_3 \end{Bmatrix}$$

It is assumed that transverse moment of inertia is included in platform inertia.

$$\vec{\omega}_r = \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \Omega \end{Bmatrix}, \quad I_r = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & C_2 \end{bmatrix},$$

$$\Rightarrow \vec{h}_r = \begin{Bmatrix} 0 \\ 0 \\ C_2\Omega \end{Bmatrix}$$

$$= \frac{1}{2} [A\omega_1^2 + A\omega_2^2 + C_1\omega_3^2 + C_2\Omega^2]$$

$$\begin{aligned}
2E_k &= A\omega_t^2 + C_1\omega_3^2 + C_2\Omega^2 \\
&= \text{const.}
\end{aligned} \tag{1}$$

Now, with energy dissipation (due to damping) kinetic energy is not constant but is a negative function of time.

$$\frac{dE_k}{dt} = A\omega_t\dot{\omega}_t + C_1\omega_3\dot{\omega}_3 + C_2\Omega\dot{\Omega} = \dot{E}_{k1} + \dot{E}_{k2} < 0 \tag{2}$$

For zero external torque, the angular momentum is constant

$$\dot{h} = 0$$

$$h^2 = (A\omega_t)^2 + (C_1\omega_3 + C_2\Omega)^2$$

$$\frac{d h^2}{dt} = 2h\dot{h} = 2A^2\omega_t\dot{\omega}_t + 2(C_1\omega_3 + C_2\Omega)(C_1\dot{\omega}_3 + C_2\dot{\Omega})$$

$$\dot{h} = \frac{A^2\omega_t\dot{\omega}_t + (C_1\omega_3 + C_2\Omega)(C_1\dot{\omega}_3 + C_2\dot{\Omega})}{h} = 0$$

$$A^2\omega_t\dot{\omega}_t + (C_1\omega_3 + C_2\Omega)(C_1\dot{\omega}_3 + C_2\dot{\Omega}) = 0 \tag{3}$$

Eq. (2) and Eq. (3) must be satisfied simultaneously

$$Eq.(3) \rightarrow A\omega_t\dot{\omega}_t = -\frac{1}{A}[(C_1\omega_3 + C_2\Omega)(C_1\dot{\omega}_3 + C_2\dot{\Omega})]$$

$$\begin{aligned}
Eq.(2) &\rightarrow -\frac{1}{A}(C_1\omega_3 + C_2\Omega)C_1\dot{\omega}_3 - \frac{1}{A}(C_1\omega_3 + C_2\Omega)C_2\dot{\Omega} + C_1\omega_3\dot{\omega}_3 + C_2\Omega\dot{\Omega} \\
&= \left[-\frac{C_1\omega_3 + C_2\Omega}{A} + \omega_3 \right] C_1\dot{\omega}_3 + \left[-\frac{C_1\omega_3 + C_2\Omega}{A} + \Omega \right] C_2\dot{\Omega} \\
&= \dot{E}_{k1} + \dot{E}_{k2}
\end{aligned}$$

$$\dot{E}_{k1} + \dot{E}_{k2} = -\lambda_1 C_1\dot{\omega}_3 - \lambda_2 C_2\dot{\Omega}$$

$$\text{where } \lambda_1 = \frac{C_1\omega_3 + C_2\Omega}{A} - \omega_3$$

$$\lambda_2 = \frac{C_1\omega_3 + C_2\Omega}{A} - \Omega$$

$$\begin{aligned} \Rightarrow C_1 \dot{\omega}_3 &= -\frac{\dot{E}_{k1}}{\lambda_1} \quad ; \text{ for platform} \\ C_2 \dot{\Omega} &= -\frac{\dot{E}_{k2}}{\lambda_2} \quad ; \text{ for rotor} \end{aligned} \quad (4)$$

Substituting eq. (4) into eq. (3) yields,

$$A \omega_t \dot{\omega}_t = \lambda_0 \left(\frac{\dot{E}_{k1}}{\lambda_1} + \frac{\dot{E}_{k2}}{\lambda_2} \right) \quad (5)$$

$$\begin{aligned} \text{where } \lambda_0 &= \frac{C_1 \omega_3 + C_2 \Omega}{A} \\ &= \frac{h_3}{A} > 0 \quad \text{total system nutation frequency (precession)} \end{aligned}$$

eq. (5) shows that $\dot{\omega}_t < 0$ (θ decreases) when

$$\frac{\dot{E}_{k1}}{\lambda_1} + \frac{\dot{E}_{k2}}{\lambda_2} < 0 \quad ; \text{ useful stability criterion} \quad (6)$$

For a satellite with an essentially despun platform, ω_3 is equal to the orbital rate or negligibly small for inertially fixed instruments.

Thus,

$$\begin{aligned} \lambda_1 &\approx \frac{C_2 \Omega}{A} \\ \lambda_2 &\approx \frac{C_2 \Omega}{A} - \Omega \\ &= \left(\frac{C_2}{A} - 1 \right) \Omega \end{aligned}$$

The stability criterion now becomes,

$$\frac{\dot{E}_{k1}}{\frac{C_2}{A}} + \frac{\dot{E}_{k2}}{\left(\frac{C_2}{A} - 1 \right)} < 0 \quad (7)$$

1) Therefore, for a satellite with spin moment of inertia $C_2 > A$, the stability criterion is always satisfied (because $\dot{E}_{k1} < 0, \dot{E}_{k2} < 0$).

This means that a damper can be placed on either body to damp out nutation.

$C_2 > A$; oblate rotor

2) when $C_2 < A$; prolate rotor

$$\frac{\dot{E}_{k2}}{\left(\frac{C_2}{A} - 1\right)} > 0 \quad (\because \dot{E}_{k2} < 0, \text{ energy dissipation on the rotor})$$

$$\rightarrow \frac{\dot{E}_{k1}}{\frac{C_2}{A}} \text{ must be larger negative quantity for stability}$$

This can be achieved by placing a large energy dissipater(damper) on the despun part ($\dot{E}_{k1} < 0$) to overcome the destabilizing damper on the spinning part.

$$\left| \frac{\dot{E}_{k1}}{\frac{C_2}{A}} \right| \gg \left| \frac{\dot{E}_{k2}}{\left(\frac{C_2}{A} - 1\right)} \right|$$

$$\rightarrow \left| \dot{E}_{k1} \right| \gg \left| \dot{E}_{k2} \left(\frac{1}{1 - \frac{A}{C_2}} \right) \right|$$

Damper를 platform에 장착한다.!!

The platform must dissipate energy at a rate for greater than the rotor to ensure that any disturbance will die out.

For prolate dual-spinners, stability is ensured if the platform dissipates energy must faster than the rotor. Any conning motion will die out.

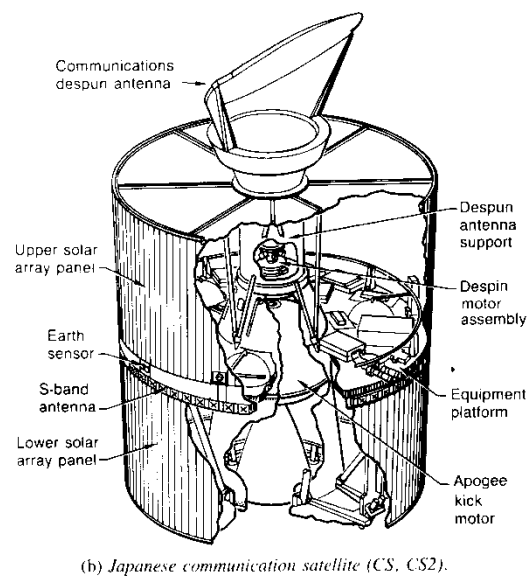
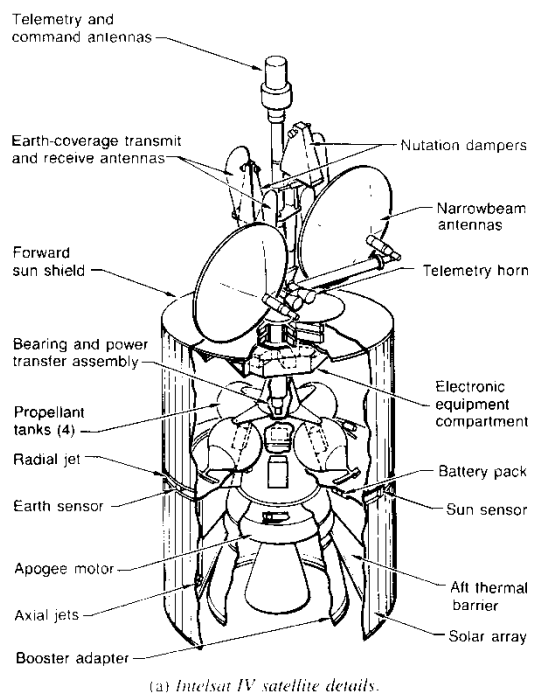


Figure 3.2 Dual-spin stabilization examples.